Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/18/2013
PI Name:	Dulchavsky, Scott A. M.D., Ph.D.		
Project Title:	Microgravity Associated Compartmental Equilibration (MACE)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Detroit	**State:**	MI
Zip Code:	48202-2608	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	10/10/2012	**End Date:**	04/09/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Villarreal, Jennifer	**Contact Phone:**	281-483-7306
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Flight Program:	ISS		
Flight Assignment:	Flight Definition		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Ebert, Douglas (Wyle Laboratories, Inc.) Garcia, Kathleen (Wyle Laboratories, Inc.) Hamilton, Douglas (Wyle Laboratories, Inc.) Killu, Keith (Henry Ford Health System) Sargsyan, Ashot (Wyle Laboratories, Inc.)		
Grant/Contract No.:	NNX13AB42G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE this project continues as "Fluid Shifts" WITH THE SAME PRINCIPAL INVESTIGATOR, Dr. Scott Dulchavsky; PERIOD OF PERFORMANCE IS 5/16/2013-9/30/2018. SEE THAT PROJECT FOR SUBSEQUENT REPORTING. (Ed., 6/18/13) Recent evidence suggests that a preexisting subclinical pathology or risk factor may determine whether microgravity fluid shifts alone or in combination with other factors causes pathological increases in intracranial pressure (ICP) and sequelae thereof (visual acuity changes, cognitive impairment, retinal pathology). Furthermore, this malady may persist in some crewmembers beyond the rehabilitation phase after flight. Space medicine is therefore facing an occupational medical hazard, with a number of contributing factors challenging to quantify and examine independently (i.e., resistive exercise vs. microgravity fluid shifts). Over years, the U.S. and the Russian space programs have implemented extensive research to understand the alterations in neurological and cardiovascular physiology secondary to fluid shifts that are induced by microgravity. The investigator team proposes to meet, through this study, the challenge of "how do we understand the effect of fluid shift on cerebrospinal fluid (CSF) dynamics when gold-standard terrestrial technologies are invasive and the true magnitude of microgravity fluid shifts and their pathophysiological effects are not well understood?" Recent on-orbit long duration studies performed by this team have shown that cephalad venous fluid shifts are permanent in all crewmembers and that external jugular pressures are increased, so that aggressive intra-thoracic pressure maneuvers (i.e., Mueller maneuver) have very little effect on extracranial venous dimensions in all crewmembers examined in space (n=6). Preliminary data from the Braslet Investigation Grant ground experiment (ongoing) find terrestrial supine position to be very different in this regard. In microgravity, cephalad fluid shifts seem to cause a relative extra-cranial (and therefore intracranial) venous insufficiency and subsequent lymphedema (facial puffiness, loss of olfactory and gustatory sense); several crew members have anecdotally mentioned that Russian thigh venous occlusion cuffs (Braslet, Kentavr nauka, Russian Federation) improve these changes and "help think clearer and perform procedures better." In our experience, venous insufficiency indeed improved when the cuffs were worn for at least 15 minutes: Mueller maneuver was much more readily causing a decrease in jugular venous dimension; cardiac preload was significantly diminishing when measured with echocardiography on-orbit. The venous thigh occlusion cuffs therefore, though altering the fluid shifts and reducing venous insufficiency, also may effect a change in CSF dynamics in space (that would explain the improvement in cognition and well-being). The investigators plan to use multiple non-invasive technologies to assess fluid shifts quantitatively, with a commensurate focus on the likelihood and severity of ocular and CSF fluid dynamic changes in microgravity, and identify markers of susceptibility to the development of pathological changes. Research questions/ Specific Aims: 1. Determining the magnitude and time course of fluid in continuous microgravity. 2. Determining the physiological and potential health effects of microgravity-associated fluid redistribution on the cardiovascular and central nervous (CNS) systems. 3. Assessing the acute fluid volumetric effects of the Braslet and Lower Body Negative Pressure (LBNP) countermeasures on central, peripheral vascular systems and CNS/ocular systems. 4. Determining best combinations of non-invasive technologies related to fluid shifts and ICP elevation that are most predictive in estimating the absolute ICP and trends thereof during space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will evaluate non-invasive measurement of fluid compartments using novel technologies including ultrasound. These techniques, if verified, would provide a significant benefit to medical care on Earth, especially in rural, remote, or underserved locations.
Task Progress:	NOTE that project continues as "Fluid Shifts" WITH THE SAME PRINCIPAL INVESTIGATOR; PERIOD OF PERFORMANCE IS 5/16/2013-9/30/2018. SEE THAT PROJECT FOR SUBSEQUENT REPORTING. (Ed., 6/18/13) New project for FY2013 in November 2012.
Bibliography Type:	Description: (Last Updated: 07/24/2025)